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import pandas as pd
data = pd.read_excel("games.csv.xlsx")
concepts = data.iloc[:, :-1].values
target = data.iloc[:, -1].values
S = concepts[0].copy()
G = [["?" for _ in range(len(S))] for _ in range(len(S))]
print("Initial Specific Hypothesis:", S)
print("Initial General Hypothesis:", G)
for i, h in enumerate(concepts):
    print("\nTraining Example", i + 1, ":", h, "->", target[i])
    if target[i] == "Yes":
        for j in range(len(S)):
            if h[j] != S[j]:
                S[j] = "?"
                G[j][j] = "?"
    else:
        for j in range(len(S)):
            if h[j] != S[j]:
                G[j][j] = S[j]
            else:
                G[j][j] = "?"
    print("Specific Hypothesis (S):", S)
    print("General Hypothesis (G):")
    for g in G:
        print(g)
print("\nFinal Specific Hypothesis:", S)
print("\nFinal General Hypothesis:")
for g in G:
    print(g)

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... Training Example 4 : ['Sunny' 'Warm' 'High' 'Strong' 'Cool' 'Change'] -> Yes
Specific Hypothesis (S): ['Sunny' 'Warm' '?' 'Strong' '?' '?']
General Hypothesis (G):
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']

Final Specific Hypothesis: ['Sunny' 'Warm' '?' 'Strong' '?' '?']

Final General Hypothesis:
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']

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